Phase 3 Report: Sentence-Role Classification in Medical Abstracts

Bio_ClinicalBERT vs. Classical ML on PubMed 20k RCT

Project F26-07 | Track A (Research & Development) | AI2002 Artificial Intelligence | Instructor: Hajra Waheed

Azan Wasty (24L-2557), Nauman Iqbal (24L-2565), Afras Shahnawaz (24L-2547)

1. Overview

This project classifies the rhetorical role of a single sentence from a medical abstract as one of five classes: BACKGROUND, OBJECTIVE, METHODS, RESULTS or CONCLUSIONS. It is a supervised, sentence-level, five-class text classification task on the PubMed 20k RCT dataset (Dernoncourt and Lee, 2017). Our Track A question is how much a fine-tuned clinical transformer (Bio_ClinicalBERT) improves over classical machine learning on TF-IDF text features, whether the difference is statistically significant, and what it costs in training and inference time.

Phase 3 delivers the full pipeline for both model families, as stated in our proposal: Logistic Regression, Random Forest and XGBoost on TF-IDF of the sentence text, and Bio_ClinicalBERT fine-tuned with the HuggingFace Trainer. All main comparisons use **sentence text only**, because the proposal places sentence position and neighbouring sentences out of scope. Data collection and EDA were completed in Phase 2 and are not repeated here. The data are the official splits: 180,040 training sentences (15,000 abstracts), 30,212 dev sentences (2,500 abstracts) and 30,135 test sentences (2,500 abstracts). Training class shares are METHODS 33.0%, RESULTS 32.2%, CONCLUSIONS 15.1%, BACKGROUND 12.1% and OBJECTIVE 7.7%, so macro-F1 is our primary metric.

2. Model Selection and Justification

Classical family (TF-IDF + linear/tree models). TF-IDF with word 1-2 grams (lowercased, minimum document frequency 3, sublinear term frequency) is the standard strong baseline for text classification and shows what lexical cues alone can achieve. We use three models from the proposal: **Logistic Regression** (log-loss trained with SGD, because a full solver is very slow on 200,000 sparse features; 200k-feature TF-IDF), **Random Forest** (100 trees) and **XGBoost** (300 trees, depth 6, GPU histogram method). The two tree models use a reduced 20,000-feature TF-IDF because trees cannot exploit 200k sparse columns efficiently. All three use balanced class weights.

Transformer family. We fine-tune `emilyalsentzer/Bio_ClinicalBERT` (Alsentzer et al., 2019), a BERT model further pretrained on clinical text (108.3M parameters), with a classification head on the [CLS] token. Its tokenizer averages 1.38 subword tokens per word on our sentences. Like the classical models, it receives the sentence text only.

3. Implementation Details and Code Workflow

Everything runs in one Jupyter notebook (F26-07.ipynb) on Google Colab with a free Tesla T4 GPU, using pandas, NumPy, scikit-learn, SciPy, XGBoost, PyTorch and HuggingFace Transformers. All random components use seed 42. The pipeline is raw .txt files, parser, DataFrame, TF-IDF or tokenizer, model, predictions, metrics and significance tests:

- **Parse and check.** Each file has ###ID lines followed by LABEL<TAB>sentence lines. The parser keeps the last abstract in a file even without a trailing blank line. Checks confirm no unknown labels and no abstract shared between splits (no leakage). The training set has 1,158 duplicate sentences, which we kept.
- **Light cleaning.** Whitespace is normalised. Numbers, punctuation and stopwords are kept, since numbers cue RESULTS and tense words such as "was" and "will" separate METHODS from RESULTS (as decided in Phase 2).
- **Classical branch.** TF-IDF is fit on the training set only, then applied to dev and test.
- **Transformer branch.** Sentences are tokenised (maximum 128 tokens, dynamic padding per batch) and the model is fine-tuned on all 180,040 training sentences.
- **Evaluation.** Accuracy, weighted F1, macro-F1 and Cohen's kappa on dev and test; confusion matrices; error analysis; McNemar's exact test and a paired abstract-level bootstrap for significance. Models are selected on dev and reported on test.

4. Training Process

Data split. We use the official PubMed 20k RCT splits: train 180,040 sentences (15,000 abstracts), dev 30,212 (2,500 abstracts) and test 30,135 (2,500 abstracts). Abstracts never cross splits. TF-IDF vocabularies and all model fitting use the training set only; dev is used for epoch and model selection; test is used once for final reporting. No cross-validation was used.

Classical setup. Logistic Regression: SGD, log-loss, alpha 2e-6, 30 iterations. Random Forest: 100 trees, sqrt features per split, minimum leaf size 3, balanced-subsample weights. XGBoost: 300 trees, depth 6, learning rate 0.2, subsample 0.8, column sample 0.5, balanced sample weights.

Bio_ClinicalBERT setup (as in the proposal). Maximum length 128, batch size 32, **5 epochs**, learning rate 2e-5 with linear decay, 6% warmup (1,688 of 28,135 steps), weight decay 0.01, fp16 on the T4, best checkpoint chosen by dev macro-F1. Training used the full training set and took 4,999.7 s (83 minutes).

Epoch	Train loss	Dev loss	Dev accuracy	Dev macro-F1
1	0.3676	0.3312	0.8783	0.8189
2	0.2986	0.3391	0.8801	0.8238
3	**0.2389**	**0.3664**	**0.8798**	**0.8253**
4	0.1799	0.4169	0.8777	0.8233
5	0.1522	0.4699	0.8719	0.8170

Table 1: Bio_ClinicalBERT per-epoch training. Dev loss is lowest after epoch 1 and rises afterwards while training loss keeps falling, a sign of overfitting. Dev macro-F1 peaks at epoch 3, so the epoch-3 checkpoint is the one evaluated on test.

5. Results

All scores are on the test set (30,135 sentences) from a single run. Rows 1-4 are the text-only models from the proposal; the last row is an optional extension (Section 5.4) and is not part of the proposed comparison.

Model	Acc.	Weighted F1	Macro-F1	Kappa	Dev macro-F1	Train time	Infer. (s/1k sent.)
Bio_ClinicalBERT [text]	**0.8698**	**0.8689**	**0.8143**	**0.8247**	**0.8253**	**4,999.7 s**	**1.656 (GPU)**
LogReg (SGD) [text]	0.8236	0.8232	0.7630	0.7630	0.7733	9.3 s	0.073 (CPU)
XGBoost [text]	0.7862	0.7891	0.7305	0.7149	0.7392	78.6 s	0.045 (CPU)
Random Forest [text]	0.7722	0.7726	0.7135	0.6946	0.7230	159.8 s	0.067 (CPU)
LogReg + position (extension)	0.8792	0.8788	0.8295	0.8378	0.8329	11.7 s	0.073 (CPU)

Table 2: Test-set results. Inference time covers feature extraction or tokenisation plus prediction on the full test set. Classical models ran on CPU and Bio_ClinicalBERT on GPU, so times are not like-for-like hardware.

5.1 RQ1: accuracy. The best classical model, chosen on dev, is Logistic Regression (macro-F1 0.763), ahead of XGBoost (0.731) and Random Forest (0.714). Bio_ClinicalBERT reaches 0.814, a gain of 5.1 macro-F1 points over the best classical model, and also leads on accuracy (0.870 vs 0.824) and kappa (0.825 vs 0.763). The Logistic Regression coefficients are interpretable and reflect the task: top terms for RESULTS include "respectively", "showed" and "significantly"; for OBJECTIVE "whether", "evaluate", "investigate" and "examine"; for CONCLUSIONS "should", "further" and "future".

Class	LogReg F1	Bio_ClinicalBERT P	Bio_ClinicalBERT R	Bio_ClinicalBERT F1	Support
BACKGROUND	0.655	0.701	0.752	0.726	3,621
OBJECTIVE	0.614	0.699	0.614	0.654	2,333
METHODS	0.904	0.927	0.950	0.938	9,897
RESULTS	0.885	0.917	0.923	0.920	9,713
CONCLUSIONS	0.757	0.863	0.807	0.834	4,571

Table 3: Per-class test results. Bio_ClinicalBERT is better on every class; the largest gains are on CONCLUSIONS (+7.7 F1 points) and BACKGROUND (+7.1).

5.2 RQ2: statistical significance. We compare Bio_ClinicalBERT with each classical model on the test predictions using McNemar's exact test and a paired bootstrap (1,000 resamples of whole abstracts, since sentences within an abstract are not independent) of the macro-F1 difference. With three comparisons, the Bonferroni threshold is 0.05/3 = 0.0167.

Classical model (A)	Macro-F1 diff (BERT - A)	95% CI (abstract bootstrap)	Only A right	Only BERT right	McNemar p
LogReg	+0.0512	[+0.0456, +0.0570]	1,046	2,439	2.3e-126
XGBoost	+0.0838	[+0.0775, +0.0898]	1,092	3,611	5.0e-311
Random Forest	+0.1008	[+0.0943, +0.1070]	1,078	4,020	below 1e-300

Table 4: Significance tests. All three differences are significant at the Bonferroni threshold and every confidence interval excludes zero, so Bio_ClinicalBERT's advantage over text-only classical models is not due to chance. This reflects test-set sampling uncertainty only; training used a single seed.

5.3 RQ3: accuracy versus efficiency. The accuracy gain is expensive. Bio_ClinicalBERT trained in about 83 minutes versus 9.3 seconds for Logistic Regression (roughly 540 times longer) and needs a GPU. Its inference is about 23 times slower per 1,000 sentences (1.66 s on GPU vs 0.07 s on CPU) and the model has 108.3M parameters. Logistic Regression gives about 94% of Bio_ClinicalBERT's macro-F1 (0.763 vs 0.814) at a tiny fraction of the cost, which makes it the better choice when compute or latency is limited. The tree models were both slower and less accurate than Logistic Regression on sparse TF-IDF text.

5.4 Extension: effect of sentence position (outside proposal scope). To see how much of the gap is a feature gap rather than a model gap, we added position features (position decile, first and last sentence flags, a number flag and log length) to Logistic Regression only. It reaches macro-F1 0.830, slightly above text-only Bio_ClinicalBERT (0.814). Part of the transformer's difficulty is therefore missing context: concluding sentences often read like results, and position is what separates them. This extension is not used in the significance tests, and giving the transformer the same information is future work.

6. Error Analysis

Bio_ClinicalBERT misclassified 3,923 of 30,135 test sentences (13.0%). The largest confusions are OBJECTIVE predicted as BACKGROUND (732), BACKGROUND predicted as OBJECTIVE (528), CONCLUSIONS predicted as RESULTS (493), RESULTS predicted as METHODS (442), CONCLUSIONS predicted as BACKGROUND (313) and METHODS predicted as RESULTS (295). OBJECTIVE is the hardest class for both families: 31% of true OBJECTIVE sentences are predicted as BACKGROUND by the transformer, matching the overlap we expected from Phase 2. Accuracy by position (Figure 1, right) shows that Bio_ClinicalBERT beats Logistic Regression at every position, and that both models are weakest at the start of the abstract (first decile: 0.731 vs 0.670), where BACKGROUND and OBJECTIVE sentences overlap, and near the end (last decile: 0.876 vs 0.838).

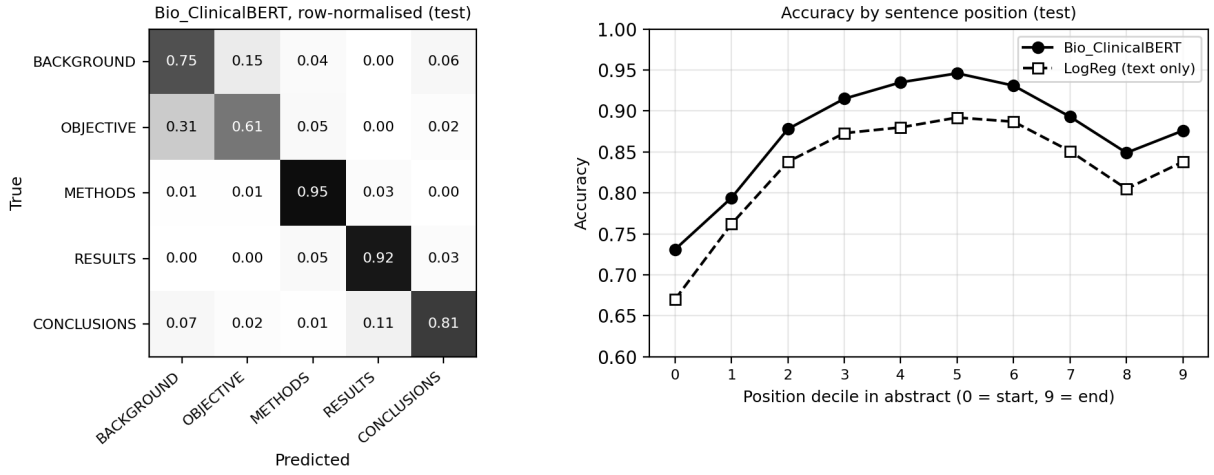


Figure 1: Left: row-normalised confusion matrix of Bio_ClinicalBERT on the test set. Right: test accuracy by position decile for Bio_ClinicalBERT and text-only Logistic Regression (position is a diagnostic here, not a model input).

Inspecting misclassified examples shows label ambiguity and noise: some aims are labelled BACKGROUND, trial registration numbers appear under BACKGROUND, and several concluding sentences describe findings in the same wording as RESULTS sentences. This sets a ceiling on achievable accuracy, especially for OBJECTIVE.

7. Challenges Faced

- **Slow logistic regression.** A full solver was too slow on 200,000 sparse features, so we used SGD with log-loss (about 9 seconds). A convergence warning appears but does not noticeably change the scores.
- **Cost of the transformer.** The proposed 5 epochs on all 180k sentences took 83 minutes on a free T4 GPU, and the data and model downloads need a stable Colab session. Checkpoints are saved to Google Drive each epoch to survive disconnects.
- **Overfitting.** Dev loss rose after epoch 1 and macro-F1 peaked at epoch 3, so we select the checkpoint by dev macro-F1 rather than using the last epoch.
- **Library versions.** The installed Transformers version rejected the optional `group_by_length` argument, so we removed it and kept the plain proposed setup. XGBoost also printed a device-mismatch warning (GPU model, CPU input) that only affects speed, not results.
- **Label ambiguity.** Noisy BACKGROUND/OBJECTIVE labels limit what any model can reach.

8. Deviations, Limitations and Next Steps

- **Deviations from the proposal.** Random Forest and XGBoost use a 20k-feature TF-IDF while Logistic Regression uses 200k, and Logistic Regression is trained with SGD, both for speed. The position extension is an add-on, not part of the proposed scope. The optional second transformer (BERT/SciBERT) comparison and the full project package structure are not part of this notebook-only Phase 3.
- **Single run.** All numbers come from one seed (42). The significance tests measure test-set sampling uncertainty, not training-seed variance.
- **Hardware.** Classical models ran on CPU and the transformer on GPU, so efficiency numbers are indicative, not hardware-matched.
- **Next steps (Phase 4).** Repeat training with several seeds, add a bias and error analysis by class and sentence length (as required for Track A), give the transformer position and neighbouring-sentence context for a like-for-like comparison with the position extension, and add a general-domain or biomedical comparison model if time allows.
- **Scope.** Results are for published RCT abstracts and do not transfer directly to clinical notes.

References

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